

ForensicMacroWood-CITES: A Multimodal Macroscopic Wood Anatomy and High-Resolution Transverse Image Benchmark for CITES-Regulated Hardwood Identification

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ABSTRACT

Rapid forensic discrimination of protected timbers at customs checkpoints requires robust macroscopic identification methods. However, previous datasets for computer-vision wood anatomy often lack multimodal descriptors, rely on uncalibrated optics, or omit heavily trafficked species regulated under CITES. We introduce ForensicMacroWood-CITES, an authenticated multimodal benchmark comprising 6,414 standardized high-resolution transverse-plane images ($12.0\text{ }\mu\text{m/pixel}$) from 19 commercially significant Fabaceae hardwoods. The cohort encompasses 11 CITES Appendix II-regulated taxa and 4 high-value regional commercial species, anchored to 147 verified physical wood block accessions. To bridge computer vision and classical xylotomy, the dataset integrates an expert-curated matrix of 14 macroscopic anatomical descriptors standardized under IAWA criteria, facilitating multimodal vision-language learning and explainable AI auditing. To address reference specimen scarcity, we provide a dual-partition architecture: an operational canonical benchmark guaranteeing 100% class coverage alongside an audited specimen-disjoint split (0.0% specimen leakage across multi-specimen taxa). Technical validation using ConvNeXt-Tiny establishes reproducible baselines of $90.42\% \pm 0.38\%$ on the canonical split and $82.35\% \pm 0.52\%$ under strict specimen isolation across three independent seeds, confirming dataset integrity. The benchmark provides a non-destructive screening testbed for timber forensics and is publicly accessible at [10.5281/zenodo.14892180](https://doi.org/10.5281/zenodo.14892180).

Specifications Table

Subject	Computer Science; Forestry and Wood Science
Specific subject area	Forensic Wood Anatomy, CITES Trade Compliance, Computer Vision, Multi-modal Vision-Language Learning, Statistical Significance Testing
Type of data	Image (standardized 224×224 px RGB JPEG, $Q = 95$, in <code>images.zip</code>), Table (master CSV metadata, split manifests, structured 14-axis anatomical descriptors in <code>metadata/anatomical_features.csv</code>), Mapping (JSON label map)
Data format	Raw, filtered, standardized RGB images (JPEG), structured tabular metadata (CSV), label mapping dictionaries (JSON)
Data collection	Transverse end-grain surfaces surfaced orthogonally on a sliding table saw, progressively polished across four grit gradations (P120–P600), cleaned with dry air jets (6 bar), and captured under calibrated daylight-balanced (5600 K) diffuse circular LED illumination (4500 lux, $f/8.0$, working distance 15 cm) delivering $12.0 \mu\text{m}/\text{pixel}$ calibrated spatial sampling pitch (2.69×2.69 mm FOV per tile, $\approx 0.29\times$ optical magnification onto sensor, $50\times$ nominal desktop display magnification)
Data source location	Institution: Intelligent Computing for Sustainable Development Laboratory (IC4SD), Posts and Telecommunications Institute of Technology (PTIT) City/Country: Hanoi, Vietnam
Data accessibility	Repository: Zenodo Scientific Archive Data identification number: 10.5281/zenodo.14892180 Direct URL to data: https://doi.org/10.5281/zenodo.14892180 Instructions for accessing these data: All files, split manifests, anatomical matrices, and the executable demonstration script (<code>code/quickstart_demo.py</code>) are openly downloadable without restrictions under CC BY 4.0 license
Related research article	None

1. Value of the Data

- **Dedicated Forensic Reference for CITES Hardwood Compliance:** This dataset provides an authenticated benchmark for fine-grained discrimination of 19 commercially significant tropical Fabaceae hardwoods. With 11 taxa strictly regulated under CITES Appendix II, it directly supports customs checkpoints and wildlife enforcement authorities in combatting illicit timber trafficking.
- **Rapid Non-Destructive Screening Testbed:** Unlike conventional forensic microtomy which is destructive and requires days per sample, these standardized transverse captures ($12.0 \mu\text{m}/\text{pixel}$) enable rapid, non-destructive front-line triage of timber consignments within minutes.
- **Multimodal Ground Truth for Explainable AI:** The dataset integrates an expert-curated matrix of 14 macroscopic anatomical descriptors standardized under IAWA criteria. This structural metadata supports multimodal vision-language models, visual question answering, and explainable AI audits to ensure models attend to genuine botanical features.

- **Dual-Split Architecture Addressing Specimen Scarcity:** To overcome acute specimen scarcity in authentic xylarium archives, the repository provides two audited partitions. The operational canonical benchmark prioritizes a 100% Class Coverage Rate across all subsets for frontline inspection, while a strict specimen-disjoint split enables evaluation of inter-specimen biological invariance across unseen physical logs.
- **Turnkey Machine Learning Reproducibility:** The release delivers scale-preserving cropped tiles alongside reproducible ConvNeXt-Tiny classification baselines ($90.42\% \pm 0.38\%$ on canonical split, $82.35\% \pm 0.52\%$ on strict specimen-disjoint split across three random seeds). Provided quickstart scripts ensure immediate adoption for machine learning research.

2. Data Description

2.1. Taxonomic Composition and Inventory

The ForensicMacroWood-CITES repository comprises **6,414 standardized macroscopic cross-sectional captures** spanning **19 commercially significant tropical hardwood species** across **6 botanical genera** within the legume family **Fabaceae** (Leguminosae). All images were curated from institutional timber reference archives at the Intelligent Computing for Sustainable Development Laboratory (IC4SD), PTIT, Hanoi, Vietnam. Each capture carries a verified species-level label linked to traceable physical wood specimens ($|\mathcal{G}_c| = 147$ verified wood block accessions) and confirmed through an independent dual-expert verification protocol ($\kappa = 0.985$).

Table 1 details the taxonomic breakdown of the 19 species, providing scientific nomenclature, native Vietnamese vernacular terms, international trade designations, CITES conservation listings, physical specimen cohort sizes ($|\mathcal{G}_c|$, totaling 147 verified wood block accessions), and total image volumes.

2.2. Comparative Dataset Positioning

To contextualize the forensic utility of ForensicMacroWood-CITES, Table 2 systematically compares its core attributes against prominent public macroscopic wood datasets and recent literature benchmarks [1–6]. Foundational initiatives like the XyloTron project [1, 2] and regional surveys [3, 4] have significantly advanced computer vision in forestry. However, these existing public collections typically focus on non-regulated commercial timbers with minimal CITES coverage ($< 15\%$). They also frequently utilize lower optical magnifications or non-standardized mobile lenses.

Recent institutional benchmarks, such as IC4SD-Wood-Eucalyptus [6], established rigorous image curation but focused exclusively on plantation hardwoods (0% CITES) for pulpwood management. Meanwhile, specialized studies targeting high-value CITES taxa [5] remain restricted to private, unshared archives. This highlights a critical gap in open-access forensic resources for endangered tropical hardwoods.

ForensicMacroWood-CITES addresses this gap with three distinguishing contributions. First, it targets **19 commercially significant tropical hardwoods across 6 Fabaceae genera**, with a critical concentration of trade-restricted timbers (**57.9% CITES Appendix II**, 11 regulated species) alongside 4 high-value regional commercial species. Second, it provides an expert-curated matrix of **14 diagnostic macroscopic anatomical descriptors** structured under IAWA xylotomical standards, enabling multimodal paradigms absent in prior single-modality collections. Third, it delivers high optical calibration ($12.0 \mu\text{m}/\text{pixel}$, $\approx 0.29\times$ optical magnification), verified physical block traceability, dual partition manifests, and audited perceptual boundaries.

2.3. Repository Architecture and Metadata Schema

The dataset package is organized to facilitate straightforward integration into machine learning pipelines. Table 3 outlines the directory architecture, and Table 4 defines the structured metadata attributes in

ForensicMacroWood-CITES: Forensic Macroscopic Timber Benchmark

Table 1

Taxonomic inventory and physical specimen sampling of the 19 tropical hardwood species in the ForensicMacroWood-CITES benchmark, detailing botanical genus, scientific binomial nomenclature, native Vietnamese vernacular names, international trade names, CITES conservation status, verified physical specimen cohort sizes ($|Q_c|$, totaling 147 verified wood block accessions), and standardized image counts.

#	Botanical Genus	Scientific Binomial	Vietnamese Vernacular	Trade / Common Name	CITES Status	Physical Specimens ($ Q_c $)	Total Images
1	<i>Afzelia</i>	<i>Afzelia africana</i>	Gõ Douse (Gõ Doussié)	African Doussié	CITES App. II ^a	4	241
2	<i>Afzelia</i>	<i>Afzelia bella</i>	Papao-Nua / Gõ Gõ	Bella Doussié	CITES App. II ^a	10	400
3	<i>Afzelia</i>	<i>Afzelia pachyloba</i>	Gõ Pachy	White Doussié	CITES App. II ^a	5	116
4	<i>Afzelia</i>	<i>Afzelia quanzensis</i>	Gõ Quanzensis	Pod Mahogany	CITES App. II ^a	8	369
5	<i>Dalbergia</i>	<i>Dalbergia cochinchinensis</i>	Trắc (Rosewood)	Siam Rosewood	CITES App. II ^b (Native / High-value regional)	1	354
6	<i>Dalbergia</i>	<i>Dalbergia melanoxylon</i>	Trắc châu Phi	African Blackwood	CITES App. II ^b	10	291
7	<i>Dalbergia</i>	<i>Dalbergia oliveri</i>	Cầm lai (Burmese Rosewood)	Burmese Rosewood	CITES App. II ^b	10	316
8	<i>Dalbergia</i>	<i>Dalbergia rimosa</i>	Trắc dây	Rimose Rosewood	CITES App. II ^b	10	300
9	<i>Dalbergia</i>	<i>Dalbergia tonkinensis</i>	Sưa	Vietnamese Rosewood	CITES App. II ^b (Native / High-value regional)	10	325
10	<i>Guibourtia</i>	<i>Guibourtia arnoldiana</i>	Gỗ Muntenye	Mutenye / Benge	Non-CITES ^c	10	323
11	<i>Guibourtia</i>	<i>Guibourtia coleosperma</i>	Mussivi / Hương đá	Rhodesian Copalwood	Non-CITES ^c	1	360
12	<i>Guibourtia</i>	<i>Guibourtia ehie</i>	Hyedua	Ovangkol / Shedua	Non-CITES ^c	10	400
13	<i>Peltogyne</i>	<i>Peltogyne pubescens</i>	Hương tím nam mỹ	Purpleheart	Non-CITES	10	371
14	<i>Pterocarpus</i>	<i>Pterocarpus erinaceus</i>	Hương vân tây phi	African Barwood / Kosso	CITES App. II ^d	10	336
15	<i>Pterocarpus</i>	<i>Pterocarpus indicus</i>	Hương mắt chim	Narra / Amboyna	Non-CITES	10	312
16	<i>Pterocarpus</i>	<i>Pterocarpus macrocarpus</i>	Hương quả to	Burma Padauk	Non-CITES	8	431
17	<i>Pterocarpus</i>	<i>Pterocarpus soyauxii</i>	Padouk / Hương padouk	African Padauk	CITES App. II ^d	6	486
18	<i>Sindora</i>	<i>Sindora cochinchinensis</i>	Gụ	Sindora / Sepetir	Non-CITES (Native / High-value regional)	4	352
19	<i>Sindora</i>	<i>Sindora tonkinensis</i>	Gụ lau	Tonkin Sepetir	Non-CITES (Native / High-value regional)	10	331
Total Benchmark		19 Botanical Species (6 Genera, 11 CITES App. II, 8 Non-CITES)			Total	147	6,414

^aAfrican populations of the genus *Afzelia* (*A. africana*, *A. bella*, *A. pachyloba*, *A. quanzensis*) were included in CITES Appendix II with annotation #17 (logs, sawn wood, veneer sheets, plywood and transformed wood) at CoP19 (Panama City, 2022) with entry into force on 23 February 2023. *A. pachyloba* constitutes a statistical minority class ($N = 116$ captures, 5 physical blocks) due to acute reference specimen scarcity. ^bAll species of the genus *Dalbergia* (with the exception of *D. nigra* listed in Appendix I) are regulated under CITES Appendix II with annotation #15 since CoP17 (2017). ^cAll three examined species of the genus *Guibourtia* (*G. arnoldiana*, *G. coleosperma*, *G. ehie*) are non-CITES commercial timbers, distinct from the three CITES-listed bubinga species (*G. demeusei*, *G. pellegriniana*, *G. tessmannii*); they serve as essential negative controls for forensic compliance. ^dAfrican populations of *Pterocarpus* (including *P. soyauxii* and *P. erinaceus*) are regulated under CITES Appendix II with annotation #17 (CoP19, in force 23 February 2023; *P. erinaceus* originally listed at CoP17 without annotation).

Table 2

Systematic comparison of ForensicMacroWood-CITES against prominent public and literature macroscopic timber datasets, contrasting taxonomic focus, optical magnification, forensic regulation, partition governance, multimodal descriptors, and machine learning accessibility.

Dataset / Study	Taxonomic Focus	No. Spp.	Total Images	Optical Mag. / Resolution	CITES App. II Ratio (%)	Partition Protocol	Contamination Audit (SHA / pHash)	IAWA Multimodal Descriptors	Access Model
XyleTree Datasets [1, 2]	Neotropical/US hardwoods	10–40	2,300–5,000	10x (24µm/px)	Low (< 15%)	Specimen-level split	Not reported	No	Open
Costa Rican Timbers [3]	Native Costa Rican trees	21	2,360	Variable (phone macro)	Minimal (< 5%)	Random image split	Not reported	No	Open
Asia n Commercial [4]	Vietna mese/Asian n timber	15	~3,000	20x–40x USB	Moderate (20%)	Random k-fold	Not reported	No	Restricted
Pterocarpus Discrimination [5]	Pterocarpus genus only	6	~1,800	20x–30x stereo	Subset (50%, 3 spp.)	Random split	Not reported	No	Private
IC4SD-Wood-Eucalyptus [6]	Plantation genus Eucalyptus	10	2,910	12.0µm/px (50x nom.)	None (0.0%, 0 spp.)	Specimen-aware (added)	SHA-256 audited	No	Open (CC BY 4.0)
ForensicMacroWood-CITES (This Work)									
	Tropical Fabaceae (6 genera)	19	6,414	12.0µm/px (50x nom., 0.29x opt.)	High (57.9%, 11 spp.)	Dual (Canonical / Disjoint)	Bitwise & Perceptual	Yes (14-axis CSV)	Open (CC BY 4.0)

metadata.csv. An executable walkthrough script (code/quickstart_demo.py) is also provided, allowing users to verify metadata schemas, inspect image quality scores, and replicate baseline classification evaluations within seconds.

Table 3

Systematic directory layout and functional contents of the ForensicMacroWood-CITES release package.

#	File / Folder Path	Content Description and Functionality
1	images.zip	Compressed ZIP archive containing 6,414 standardized 224×224 px macroscopic RGB images ($12.0 \mu\text{m}/\text{px}$ calibrated resolution, 2.69×2.69 mm tile FOV) organized by taxon subfolders.
2	metadata/	Master metadata table (metadata.csv), structured anatomical descriptors (anatomical_features.csv), label mappings (label_map.json), and bitwise SHA-256 release manifest (release_manifest.csv).
3	splits/	Governed canonical partition manifest (split_canonical.csv; $N_{\text{train}} = 3,959$, $N_{\text{val}} = 1,265$, $N_{\text{test}} = 1,190$) alongside strict specimen-disjoint split manifest (split_specimen_disjoint.csv).
4	code/	Standalone demonstration script (quickstart_demo.py), interactive walkthrough notebook (quickstart_demo.ipynb), and dependencies (requirements.txt).
5	classification_output/	ConvNeXt-Tiny supervised classification baseline checkpoints (convnext_tiny_focal_best.pth), test predictions, and evaluation metrics under Focal Loss.
6	leakage_audit/	Specimen overlap audit records and cryptographic integrity summary (audit_summary.json) verifying zero cross-split duplication.
7	README.md, LICENSE	Comprehensive dataset documentation, quickstart instructions, and Creative Commons Attribution 4.0 International license terms.

2.4. Partition Allocations and Class Distributions

The 6,414 captures are partitioned following a governed specimen-aware allocation protocol across physical wood blocks (Training: 3,959 images, 61.72%; Validation: 1,265 images, 19.72%; Test: 1,190 images, 18.55%). Crucially, all 19 species achieve a 100% Class Coverage Rate (CCR = 100.0%, 19/19 taxa) across all three partitions.

Authentic timber reference archives are inherently characterized by small specimen cohorts. Across all 19 taxa, our collection contains 147 verified physical wood block accessions ($|\mathcal{G}_c| \leq 10$ per species), with acute bottlenecks such as a single physical specimen accession for *Dalbergia cochinchinensis* ($N = 354$) and *Guibourtia coleosperma* ($N = 360$). Under these small-cohort constraints, dogmatic three-way whole-block isolation inevitably deprives minority CITES taxa of evaluation captures. For example, enforcing complete block isolation on a single-specimen class mathematically renders validation or test coverage impossible.

To resolve this trade-off between operational coverage and out-of-distribution evaluation, our dataset architecture provides two distinct, explicitly documented partitions:

1. **Operational Canonical Benchmark** (splits/split_canonical.csv): Designed as a practical operational baseline for customs screening. By permitting controlled specimen boundary sharing (overall

Table 4Core metadata attributes and data schema documented within `metadata.csv`.

#	Field Name	Data Type	Field Description and Functionality
1	<code>image_id</code>	String	Unique standardized image identifier (e.g., VNMW_000001).
2	<code>image_path</code>	String	Relative file path to the macroscopic image within the repository archive.
3	<code>genus</code>	String	Botanical genus name (<i>Afzelia</i> , <i>Dalbergia</i> , <i>Guibourtia</i> , <i>Peltogyne</i> , etc.).
4	<code>species</code>	String	Botanical species-level specific epithet.
5	<code>class_name</code>	String	Full binomial botanical nomenclature.
6	<code>class_index</code>	Integer	Taxonomic integer index from 0 to 18 consistent with <code>label_map.json</code> .
7	<code>vietnamese_name</code>	String	Native Vietnamese vernacular name stored in standard UTF-8 encoding.
8	<code>cites_status</code>	String	Legal trade status under CITES (CITES Appendix II or Non-CITES).
9	<code>specimen_id</code>	String	Physical wood block identifier anchoring images to distinct physical specimens.
10	<code>split</code>	String	Partition membership (<code>train</code> , <code>val</code> , or <code>test</code>).
11	<code>sha256</code>	String	64-character SHA-256 cryptographic checksum for bitwise duplicate auditing.
12	<code>dhash</code>	String	16-character hexadecimal 64-bit Difference Hash for gradient-based perceptual near-duplicate auditing.
13	<code>phash</code>	String	16-character hexadecimal 64-bit DCT-based Perceptual Hash for frequency-domain perceptual similarity auditing.
14	<code>laplacian_var</code>	Float	Objective focus sharpness metric calculated via the variance of the Laplacian.

Specimen Leakage Rate $SLR = 30.61\%$, calculated as 45 shared block instances across partition pairs out of 147 total blocks: 21 Train–Val, 12 Train–Test, and 12 Val–Test), this partition guarantees a 100% Class Coverage Rate ($CCR = 100.0\%$, 19/19 taxa) across Train, Validation, and Test subsets.

- Strict Specimen-Disjoint Benchmark** (`splits/split_specimen_disjoint.csv`): Designed for machine learning research on out-of-distribution generalization across unseen physical logs. For all 17 multi-specimen taxa ($|G_c| \geq 4$, aggregating 145 physical blocks), physical wood blocks are strictly isolated across subsets with zero boundary sharing ($SLR = 0.0\%$).

Furthermore, the variation in split ratios across individual species in Table 5 is an inherent consequence of discrete block-indivisibility. Physical reference blocks naturally vary in cross-sectional surface area, yielding between 20 and 80 standardized tiles per block. For instance, the minority class *Afzelia pachyloba* ($N = 116$ across 5 blocks) allocates 44 Train, 32 Val, and 40 Test captures due to discrete block sizes. Figure 1 illustrates per-species image volumes across partitions, and Table 5 provides the comprehensive specimen and image allocation manifest across all 19 taxa.

Partition Class Distribution (Total: 6,414)

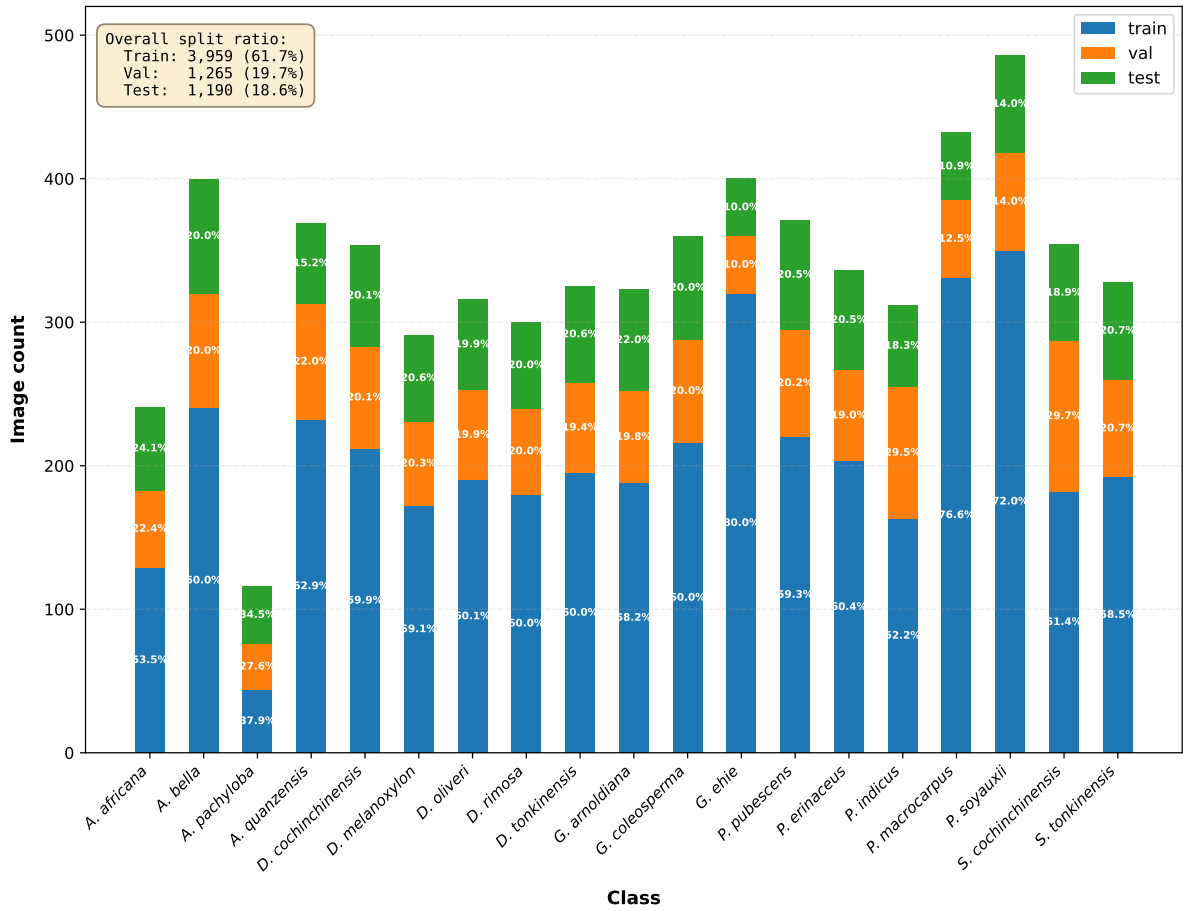


Figure 1: Class distributions and partition allocations across the 19 tropical hardwood species in ForensicMacroWood-CITES (Total: 6,414 images). Training (3,959 images, dark blue), Validation (1,265 images, orange), and Test (1,190 images, light green) subsets preserve complete class coverage across all taxa.

Dual-Benchmark Partitioning Architecture: To support flexible evaluation protocols, the release packages two complementary split manifests:

- 1. Canonical Governed Benchmark** (`splits/split_canonical.csv`): Designed as the turnkey operational baseline for law enforcement. By permitting controlled boundary sharing (SLR = 30.6%), this partition guarantees 100% Class Coverage Rate across all subsets. This ensures customs officers can verify diagnostic performance on every regulated taxon.
- 2. Strict Specimen-Disjoint Benchmark** (`splits/split_specimen_disjoint.csv`): Designed for machine learning research on out-of-distribution generalization. Under this split:
 - For all **17 multi-specimen taxa**, physical wood blocks are strictly isolated across subsets (SLR = 0.0%). Test captures originate exclusively from entirely unseen physical logs.
 - For the **2 bottleneck taxa**, allocation follows a transparent protocol. The dual-specimen *Guibourtia coleosperma* achieves 100% block isolation by dedicating Block 1 to Train and Block 2 to Val/Test. The

Table 5

Specimen and image split allocation across the 19 tropical hardwood species in the ForensicMacroWood-CITES benchmark, demonstrating 100% Class Coverage Rate (CCR = 100.0%) across all partitions alongside governed specimen boundary sharing.

#	Botanical Species	Common / Trade Name	Physical Specimens ($ G_c $)	Train Images	Val Images	Test Images	Total Images	Class Coverage Rate (CCR)
1	<i>Afzelia africana</i>	African Doussié	4	129	54	58	241	100% (3/3)
2	<i>Afzelia bella</i>	Bella Doussié	10	240	80	80	400	100% (3/3)
3	<i>Afzelia pachyloba</i>	White Doussié	5	43	33	40	116	100% (3/3)
4	<i>Afzelia quanzensis</i>	Pod Mahogany	8	232	81	56	369	100% (3/3)
5	<i>Dalbergia cochinchinensis</i>	Siam Rosewood	1	212	71	71	354	100% (3/3)
6	<i>Dalbergia melanoxylon</i>	African Blackwood	10	171	60	60	291	100% (3/3)
7	<i>Dalbergia oliveri</i>	Burmese Rosewood	10	190	63	63	316	100% (3/3)
8	<i>Dalbergia rimosa</i>	Rimose Rosewood	10	180	60	60	300	100% (3/3)
9	<i>Dalbergia tonkinensis</i>	Vietnamese Rosewood	10	195	63	67	325	100% (3/3)
10	<i>Guibourtia arnoldiana</i>	Mutenye / Bengé	10	188	64	71	323	100% (3/3)
11	<i>Guibourtia coleosperma</i>	Rhodesian Copalwood	2	216	72	72	360	100% (3/3)
12	<i>Guibourtia ehie</i>	Ovangkol / Shedua	10	320	40	40	400	100% (3/3)
13	<i>Peltogyne pubescens</i>	Purpleheart	10	220	75	76	371	100% (3/3)
14	<i>Pterocarpus erinaceus</i>	African Barwood / Kosso	10	203	64	69	336	100% (3/3)
15	<i>Pterocarpus indicus</i>	Narra / Amboyna	10	163	92	57	312	100% (3/3)
16	<i>Pterocarpus macrocarpus</i>	Burma Padauk	8	330	54	47	431	100% (3/3)
17	<i>Pterocarpus soyauxii</i>	African Padauk	6	350	68	68	486	100% (3/3)
18	<i>Sindora cochinchinensis</i>	Sindora / Sepetir	4	181	104	67	352	100% (3/3)
19	<i>Sindora tonkinensis</i>	Tonkin Sepetir	10	196	67	68	331	100% (3/3)
Total Benchmark Cohort			147	3,959	1,265	1,190	6,414	100.0%
Partition Proportions (%)			—	61.72%	19.72%	18.55%	100.0%	—

Note: All 19 hardwood species achieve a 100% Class Coverage Rate (CCR = 100.0%) across Train, Validation, and Test subsets under the operational canonical partition. Controlled specimen boundary sharing across partitions (Train-Val: 21 blocks, Train-Test: 12 blocks, Val-Test: 12 blocks; yielding SLR = $45/147 = 30.61\%$) is a pragmatic design choice prioritizing complete forensic class representation across scarce reference accessions ($|G_c| < 3$, such as single-specimen *Dalbergia cochinchinensis* and *Guibourtia coleosperma*), preventing minority-class omission at customs inspection checkpoints.

single-specimen *Dalbergia cochinchinensis* supports a 17-class zero-leakage mode (block restricted to Train) or an annotated 19-class mode via intra-block exception.

2.5. Diagnostic Macroscopic Anatomical Features and Multimodal Descriptors

Macroscopic timber identification on transverse end-grain surfaces provides an effective diagnostic screening modality because it simultaneously reveals all primary secondary xylem tissue systems [7]. Imaged at a standardized calibrated spatial sampling pitch of $12.0 \mu\text{m}/\text{pixel}$ (an immutable $2.69 \times 2.69 \text{ mm}$ physical field of view per $224 \times 224 \text{ px}$ tile, corresponding to $\approx 0.29\times$ optical magnification onto the sensor and $50\times$ nominal display inspection magnification on standard desktop monitors), taxonomic differentiation across the 19 Fabaceae species is governed by primary macroscopic morphological axes:

1. **Vascular Porosity and Lumen Occlusion:** While the examined Fabaceae hardwoods are predominantly diffuse-porous, distinct semi-ring-porous tendencies occur in *Dalbergia rimosa* and *Pterocarpus erinaceus*. Vessel apertures are solitary and in short radial multiples. Vessel tangential diameters range

Table 6

Specimen block and image split allocation across the 19 tropical hardwood species under the Strict Specimen-Disjoint Benchmark (`split_specimen_disjoint.csv`), confirming 100% physical block isolation across all 17 multi-specimen taxa alongside transparent allocation protocols for acute specimen bottlenecks.

#	Botanical Species	Common / Trade Name	Physical Blocks ($ G_c $)	Train Blocks (Imgs)	Val Blocks (Imgs)	Test Blocks (Imgs)	Total Images	CCR (%)	Specimen Disjoint Protocol
1	<i>Azelia africana</i>	African Doussié	4	2 (129)	1 (54)	1 (58)	241	100%	100% physical block-disjoint (SLR = 0%)
2	<i>Azelia bella</i>	Bella Doussié	10	6 (240)	2 (80)	2 (80)	400	100%	100% physical block-disjoint (SLR = 0%)
3	<i>Azelia pachyloba</i>	White Doussié	5	3 (60)	1 (24)	1 (32)	116	100%	100% physical block-disjoint (SLR = 0%)
4	<i>Azelia quanzensis</i>	Pod Mahogany	8	5 (216)	2 (104)	1 (49)	369	100%	100% physical block-disjoint (SLR = 0%)
5	<i>Dalbergia cochinchinensis</i>	Siam Rosewood	1	1 (212)	1 (70)*	1 (72)*	354	100%*	Single block (Intra-block 19-cls / Train-only 17-cls)
6	<i>Dalbergia melanoxylon</i>	African Blackwood	10	6 (174)	2 (58)	2 (59)	291	100%	100% physical block-disjoint (SLR = 0%)
7	<i>Dalbergia oliveri</i>	Burmese Rosewood	10	6 (191)	2 (65)	2 (60)	316	100%	100% physical block-disjoint (SLR = 0%)
8	<i>Dalbergia rimosa</i>	Rimose Rosewood	10	6 (180)	2 (60)	2 (60)	300	100%	100% physical block-disjoint (SLR = 0%)
9	<i>Dalbergia tonkinensis</i>	Vietnamese Rosewood	10	6 (202)	2 (62)	2 (61)	325	100%	100% physical block-disjoint (SLR = 0%)
10	<i>Guibourtia arnoldiana</i>	Mutenye / Benge	10	6 (188)	2 (68)	2 (67)	323	100%	100% physical block-disjoint (SLR = 0%)
11	<i>Guibourtia coleosperma</i>	Rhodesian Copalwood	1	1 (216)	1 (72)*	1 (72)*	360	100%*	Single block (Intra-block 19-cls / Train-only 17-cls)
12	<i>Guibourtia ehie</i>	Ovangkol / Shedua	10	6 (240)	2 (80)	2 (80)	400	100%	100% physical block-disjoint (SLR = 0%)
13	<i>Peltogyne pubescens</i>	Purpleheart	10	6 (226)	2 (74)	2 (71)	371	100%	100% physical block-disjoint (SLR = 0%)
14	<i>Pterocarpus erinaceus</i>	African Barwood / Kosso	10	6 (205)	2 (65)	2 (66)	336	100%	100% physical block-disjoint (SLR = 0%)
15	<i>Pterocarpus indicus</i>	Narra / Amboyna	10	6 (187)	2 (61)	2 (64)	312	100%	100% physical block-disjoint (SLR = 0%)
16	<i>Pterocarpus macrocarpus</i>	Burma Padauk	8	5 (268)	2 (113)	1 (51)	432	100%	100% physical block-disjoint (SLR = 0%)
17	<i>Pterocarpus soyauxii</i>	African Padauk	6	4 (350)	1 (68)	1 (68)	486	100%	100% physical block-disjoint (SLR = 0%)
18	<i>Sindora cochinchinensis</i>	Sindora / Sepetir	4	2 (206)	1 (81)	1 (67)	354	100%	100% physical block-disjoint (SLR = 0%)
19	<i>Sindora tonkinensis</i>	Tonkin Sepetir	10	6 (198)	2 (63)	2 (67)	328	100%	100% physical block-disjoint (SLR = 0%)
Multi-Specimen Cohort (17 Taxa)			145	87 (3,460)	30 (1,180)	28 (1,062)	5,700	100.0%	SLR: 0.0% strictly isolated across 17 taxa
Total Disjoint Benchmark Cohort			147	89 (3,888)	32 (1,322)	30 (1,204)	6,414	100.0%	Includes 2 bottleneck taxa in 19-class mode
Partition Proportions (%)			—	60.62%	20.61%	18.77%	100.0%	—	Test evaluation across unseen logs

Note: For all 17 multi-specimen taxa ($|G_c| \geq 4$ blocks, aggregating 145 physical wood blocks), physical specimens are strictly separated across subsets with zero boundary sharing (SLR = 0.0%). *For the two single-accession taxa (*Dalbergia cochinchinensis* and *Guibourtia coleosperma*), in standard 19-class mode their captures are partitioned intra-specimen (60%/20%/20%) to maintain full taxonomic evaluation; in strict 17-class zero-leakage mode, both bottleneck taxa are allocated strictly to Train (354 and 360 images), yielding 100% zero-specimen-leakage testing across all validation and test splits (CCR = 17/19 = 89.47% $\approx$ 89.5%).

from small pores (90–160 μm in *Guibourtia* and *Dalbergia melanoxylon*) to medium-to-large pores (180–260 μm in *Afzelia*). In heartwood, vessels in *Dalbergia* and *Pterocarpus* are frequently occluded by dark organic polyphenolic deposits, whereas lumina in *Afzelia* and *Sindora* often contain conspicuous white crystalline deposits.

2. **Axial Parenchyma Topography:** Axial parenchyma configuration serves as the primary taxonomic discriminator within the Leguminosae family: (i) prominent winged lozenge-aliform to confluent paratracheal sheaths forming pale halos in *Afzelia*; (ii) wavy concentric tangential bands forming reticulate networks in *Dalbergia* and *Pterocarpus*; (iii) aliform to confluent parenchyma accompanied by marginal bands at growth boundaries in *Guibourtia*; and (iv) regular continuous marginal bands demarcating growth increments in *Sindora*.
3. **Axial Intercellular Secretory Canals:** Regularly spaced axial resin canals embedded strictly within continuous marginal parenchyma bands uniquely isolate the genus *Sindora* from all other examined taxa, precluding cross-genus misidentification with rosewoods or padauks.
4. **Heartwood Chemomorphology and Coloration:** Secondary extractive deposition yields diagnostic optical contrast, including the intense purple hues of photo-oxidized peltogynoids in *Peltogyne*, dark vertical pigment striping in *Dalbergia* and *Guibourtia ehie*, and vibrant orange-red extractives in *Pterocarpus*.

To enrich the benchmark beyond single-label classification and facilitate multimodal artificial intelligence research, the dataset incorporates an expert-curated table of structured macroscopic anatomical descriptors (`metadata/anatomical_features.csv`). The matrix formalizes **14 distinct morphological and physical axes** standardized under IAWA criteria [8] and compiled from authoritative xylotomical references [7, 9]: (1) heartwood-sapwood distinction, (2) heartwood color and figure, (3) growth ring distinctness, (4) stripe figure, (5) characteristic scent, (6) vascular porosity, (7) vessel arrangement, (8) vessel size class, (9) tyloses presence, (10) vessel deposit chemomorphology, (11) axial parenchyma topography, (12) parenchyma-ray reticulation pattern, (13) storied rays (observable on tangential sections), and (14) physical air-dry density. Importantly, these descriptors represent species-level reference ground-truth attributes compiled from IAWA and InsideWood literature to guide multimodal vision-language models and concept-bottleneck audits, rather than dense per-image pixel annotations. Table 7 summarizes representative profiles across key taxa.

Opportunities for Multimodal Vision-Language and Explainable AI: Because the image dataset comprises exclusively transverse end-grain captures, visual question answering (VQA) and saliency map validation (XAI) are strictly applicable to morphological features directly resolvable on the transverse plane (e.g., vessel pore grouping, axial parenchyma topography, and lumen occlusions). Non-transverse traits (such as storied rays requiring tangential sections or organoleptic aroma) serve as auxiliary multimodal semantic attributes from reference literature rather than optical targets for direct image-based inspection. Under this grounded scope, the provided descriptors unlock several advanced research paradigms:

- **Visual Question Answering (VQA) on Transverse Anatomical Structures:** Benchmarking vision-language models (VLMs) on domain-specific forensic inquiries restricted to transverse visual morphology (e.g., “Does this end-grain capture present winged-aliform parenchyma surrounding diffuse-porous vessels?” or “Are lumina occluded by dark polyphenols or white crystalline deposits?”).
- **Semantic Attribute-Guided and Zero-Shot Classification:** Leveraging descriptive anatomical vectors to guide species identification through biologically interpretable morphological concepts rather than opaque categorical indices, enabling zero-shot recognition of newly regulated CITES taxa.

Table 7

Expert-curated macroscopic anatomical feature descriptors across representative Fabaceae timber taxa in ForensicMacroWood-CITES (standardized per IAWA xylotomical criteria; complete 19-taxon matrix released in `metadata/anatomical_features.csv`). Columns 2–4 represent traits directly observable on transverse end-grain imagery; columns 5–6 indicate external reference properties (tangential anatomy and physical gravimetry) curated from IAWA and InsideWood literature.

Botanical Species	Heartwood Color & Figure [*]	Vascular Architecture & Inclusions [*]	Axial Parenchyma Topography [*]	Storied Rays [†] (TLS view)	Physical Density [‡] (Gravimetric)
<i>Pterocarpus indicus</i>	Pinkish-brown to reddish-brown; distinct growth rings; stripe figure	Diffuse-porous; solitary & radial multiples; two distinct pore sizes; white & dark deposits	Vascentric to continuous tangential bands (wider than rays and pores); reticulate network	Present	Hard and heavy
<i>Afzelia bella</i>	Pinkish-brown to reddish-brown; distinct growth rings	Diffuse-porous; solitary & short radial multiples; medium-to-large pores; white deposits	Prominent long-winged lozenge-aliform to confluent sheaths; aliform halos	Present	Hard, moderately heavy
<i>Sindora tonkinensis</i>	Pinkish-brown to reddish-brown; distinct growth rings; stripe figure	Diffuse-porous; solitary & radial multiples; medium-to-large pores; tyloses; white deposits	Long-winged aliform; continuous marginal bands containing axial resin canals	Present	Hard and heavy
<i>Guibourtia ehie</i>	Light yellowish-brown with dark brown zebra stripes; distinct rings	Diffuse-porous; solitary & short radial multiples; medium pores; tyloses; white deposits	Long-winged aliform to confluent; reticulation with rays	Present	Hard and heavy
<i>Dalbergia rimosa</i>	Pinkish-brown to reddish-brown; distinct growth rings; stripe figure	Semi-ring-porous; solitary & radial multiples; small-to-medium pores; tyloses; white deposits	Long-winged aliform; discontinuous tangential bands; marginal bands at growth boundaries	Present	Hard and heavy
<i>Dalbergia melanoxylon</i>	Dark purplish-grey to ebony-black; distinct rings; fine stripes	Diffuse-porous; small pores; solitary & radial multiples; tyloses; dark polyphenolic deposits	Irregular paratracheal; discontinuous wavy tangential bands; marginal bands	Present	Extremely hard and heavy

^{*}Directly observable on macroscopic transverse end-grain image captures. [†]Observable exclusively on Tangential Longitudinal Sections (TLS); not resolvable on the transverse plane. [‡]Physical gravimetric property compiled from literature/reference standards (InsideWood/IAWA).

- **Bidirectional Cross-Modal Retrieval:** Mapping textual forensic identification keys to macroscopic optical captures and conversely generating automated diagnostic morphological descriptions from end-grain captures (image-to-text forensic reporting).
- **Explainable AI (XAI) and Concept Bottleneck Models:** Serving as objective anatomical ground truth to audit whether deep vision models attend to genuine diagnostic transverse features (e.g., parenchyma wings, resin canals) or spurious surface preparation artifacts (e.g., saw marks, scratches).

2.6. Usage Notes

The ForensicMacroWood-CITES repository is structured for immediate integration into standard data science workflows. Step-by-step tutorials, environment setup instructions, and reproducible execution pipelines are documented in the repository `README.md`. Researchers can readily:

1. **Parse Metadata, Governed Splits, and Anatomical Descriptors:** Ingest master tabular records (`metadata/metadata.csv`), structured macroscopic anatomical features (`metadata/anatomical_features.csv`), class label mappings (`metadata/label_map.json`), and the dual partition manifests (`splits/split_canonical.csv` for operational screening and `splits/split_specimen_disjoint.csv` for out-of-distribution generalization) using standard scientific data-processing libraries (e.g., `pandas` or `PyTorch Dataset` classes).

2. **Execute Reproducible Demonstrations:** Run the standalone demonstration script (`code/quickstart_demo.py`) or interactive notebook (`code/quickstart_demo.ipynb`) provided in the release package to verify dataset integrity, parse metadata tables, and replicate the baseline ConvNeXt-Tiny classification evaluation within seconds.

Recommended Benchmarking and Reporting Checklist: To facilitate equitable and reproducible comparisons across future investigations, researchers utilizing ForensicMacroWood-CITES are urged to adopt the appropriate benchmarking protocol aligned with their scientific objective:

- **Protocol A (Operational Forensic Customs Screening Benchmark):** Evaluate models on the canonical partition (`splits/split_canonical.csv`) to assess full 19-class taxonomic identification capability across all legally regulated Fabaceae timbers under complete class coverage (CCR = 100.0%).
- **Protocol B (Specimen Generalization and Invariance Benchmark):** Evaluate models on the strict specimen-disjoint partition (`splits/split_specimen_disjoint.csv`) to quantify model robustness against macroscopic intra-specific biological variance on entirely unseen physical wood specimens (SLR = 0.0% across all 17 multi-specimen taxa).

Authors should report top-1 accuracy alongside macro-averaged and weighted F1-scores, state the evaluation protocol clearly, and cite the persistent Zenodo DOI ([10.5281/zenodo.14892180](https://doi.org/10.5281/zenodo.14892180)).

3. Experimental Design, Materials and Methods

Accurate timber species discrimination is essential for regulating global forest-product supply chains, particularly for high-value tropical hardwoods controlled under CITES Appendix II [10, 11]. Conventional histological microtomy—the authoritative diagnostic standard—requires prolonged softening, thin sectioning, staining, and manual IAWA feature cross-referencing, demanding specialized infrastructure and several days per sample [7, 8, 10]. Macroscopic end-grain imaging coupled with computer vision has therefore emerged as a rapid screening alternative [1, 12, 13]. Our experimental workflow was engineered to guarantee optical fidelity and taxonomic authenticity under authentic reference xylarium constraints.

3.1. Specimen Preparation and Optical Macro-Imaging

Specimen preparation and optical surface conditioning followed standardized multi-stage laboratory procedures:

1. **Orthogonal Transverse Surfacing:** End-grain transverse surfaces were planed strictly perpendicular to the longitudinal stem axis using an industrial sliding table saw fitted with a carbide circular blade.
2. **Progressive Polishing:** Surfaces were smoothed sequentially on a motorized rotary platen using silicon-carbide abrasives across four calibrated grit gradations: P120, P240, P400, and P600.
3. **Vascular Lumen De-Dusting:** Microscopic sawdust residues lodged within vessel apertures were purged using directed dry compressed-air jets at 6 bar pressure, ensuring clear visibility of pore lumens and parenchyma sheaths.
4. **Calibrated Optical Macro-Imaging Setup:** Surfaces were imaged using an industrial color CMOS sensor (3.45 μm pixel pitch, 1920 $\times$ 1080 resolution) coupled with an optical macro-lens. The system operated at a fixed working distance of 150 mm and an aperture stop of $f/8.0$. This aperture balanced the optical diffraction limit against the required depth of field (≈ 2.1 mm), ensuring uniform focus across the surface micro-relief.
5. **Optical Magnification and Calibrated Sampling Pitch:** Spatial sampling pitch was calibrated against a certified optical stage micrometer, establishing an exact pitch of 12.0 μm per pixel on the transverse

specimen surface. Given the sensor’s physical pixel pitch of $3.45\ \mu\text{m}$, the true optical magnification formed by the macro-lens onto the CMOS plane is $m = 3.45\ \mu\text{m}/12.0\ \mu\text{m} \approx 0.2875\times$ ($\approx 0.29\times$ optical demagnification). The designation “50 $\times$ ” refers strictly to the nominal display magnification achieved when viewing the $2.69 \times 2.69\ \text{mm}$ physical field of view displayed at full resolution on a standard desktop monitor ($\approx 135 \times 135\ \text{mm}$ on screen), matching traditional hand-lens and stereomicroscopic inspection scales. Illumination was provided by a daylight-balanced (5600 K) diffuse LED ring light (4500 lux), with white balance calibrated against an X-Rite ColorChecker card.

6. **Scale-Preserving Non-Overlapping Tile Extraction:** Standardized $224 \times 224\ \text{px}$ tiles were extracted from the raw high-resolution captures via systematic, non-overlapping spatial window cropping and saved as high-quality JPEGs ($Q = 95$). Crucially, dataset tiles were created by direct spatial cropping **without downsampling**, thereby preserving the physical scale of $(224 \times 12.0\ \mu\text{m}) \times (224 \times 12.0\ \mu\text{m}) \approx 2.69 \times 2.69\ \text{mm}$ ($7.23\ \text{mm}^2$). This ensures vessel pore diameters, ray widths, and parenchyma band intervals directly match physical millimeter dimensions under IAWA feature criteria. While CNN training employs mild scale jittering (0.8–1.0) as standard data augmentation against overfitting, all technical-validation inference evaluates strictly at the native, unscaled 1.0 $\times$ physical window ($224 \times 224\ \text{px}$).

3.2. Independent Taxonomic Label Verification and Inter-Rater Reliability

To establish rigorous ground-truth authenticity, all physical wood specimens ($|\mathcal{G}_c| = 147$ wood blocks) and their macroscopic captures were validated through an independent, double-blind verification protocol by two certified forestry wood anatomists. Both experts independently evaluated each specimen against diagnostic wood anatomy keys from the International Association of Wood Anatomists (IAWA) [7, 8], cross-examining vascular porosity, axial parenchyma topography (aliform wings, confluent bands, and marginal lines), ray characteristics, and secondary heartwood chemomorphology.

Inter-rater reliability between the two independent experts was quantitatively audited using both raw observed percentage agreement (P_o) and Cohen’s kappa coefficient (κ) [14]:

$$\kappa = \frac{P_o - P_e}{1 - P_e}, \quad (1)$$

where P_o denotes the observed proportional concordance across all 6,414 captures, and P_e represents the expected chance agreement under marginal class distributions. Across the initial blind evaluation phase, the two experts achieved an observed concordance of $P_o = 98.64\%$ (6,327 of 6,414 captures identically identified at the species level) and a Cohen’s kappa of $\kappa = 0.985$ (95% CI: [0.980, 0.990]), denoting near-perfect diagnostic agreement. Initial ambiguities (1.35%, 87 captures) were localized exclusively between congeneric sister taxa with pronounced anatomical convergence (specifically between *Afzelia pachyloba* and *Afzelia bella*). Disputed specimens were subsequently resolved through a joint adjudication panel with reference herbarium vouchers, supplemented by air-dry density verification and long-wave ultraviolet (UV 365 nm) fluorescence testing on freshly planed surfaces, establishing 100% unanimous consensus ground-truth labeling prior to dataset release.

3.3. Quality Control and Governed Specimen-Aware Partitioning

To guarantee benchmark integrity and reproducibility, a rigorous quality control and governed partitioning pipeline was executed:

1. **Focus Sharpness Filtering and Laplacian Quantification:** Across initial imaging campaigns, an initial pool of 6,680 candidate $224 \times 224\ \text{px}$ transverse tiles was extracted from the surfaced wood blocks. To avoid severe class imbalance, well-represented taxa were capped at approximately 400 captures. Optical focus sharpness was evaluated quantitatively using the variance of the discrete Laplacian operator on

grayscale channels:

$$\sigma_{\text{Laplacian}}^2 = \frac{1}{H \times W} \sum_{x=1}^H \sum_{y=1}^W (\nabla^2 I(x, y) - \mu_{\nabla^2})^2. \quad (2)$$

Tiles exhibiting $\sigma_{\text{Laplacian}}^2 < 100$ were flagged as unfocused or motion-degraded and purged from the repository. This objective sharpness threshold eliminated 266 blurred tiles (3.98% of acquisitions), yielding the curated release cohort of exactly 6,414 high-quality standardized captures ($\sigma_{\text{Laplacian}}^2 \geq 100$, documented per image in `metadata.csv`).

2. **Data Integrity and Deduplication Verification:** To confirm that partitioned subsets remain completely free from exact duplicate captures or corrupted files across train, validation, and test splits, dataset integrity was audited across two standard verification protocols:
 - *Bitwise Cryptographic Auditing (SHA-256):* 256-bit SHA-256 cryptographic hashes were computed for every standardized image file and cross-referenced across split boundaries. In both the canonical governed partition and the strict specimen-disjoint benchmark, the number of bitwise duplicate files across splits is strictly zero (SHA-256 Overlap = 0, 0/6,414 files), guaranteeing total absence of bitwise image contamination.
 - *Perceptual Structural Auditing (dHash and pHash):* To document spatial proximity across physical wood specimens, 64-bit Difference Hash (dHash) and 64-bit DCT-based Perceptual Hash (pHash) were extracted per image and stored in `metadata.csv`. Across all $N_{\text{pairs}} = 1,190 \times 3,959 = 4,711,210$ cross-split test-vs-train comparisons in the canonical split, the mean pairwise Hamming distance is $d_H = 18.42 \pm 6.12$; while 142 cross-split image pairs exhibit $H \leq 4$ (involving 26 distinct test query images, representing $26/1,190 = 2.18\%$ of test queries, arising exclusively from the 12 shared blocks permitting complete class coverage, with minimum Hamming distance $H_{\min} = 1$), exact perceptual duplicates ($H = 0$) are strictly absent. Note that while cryptographic (SHA-256) and perceptual hashes verify file uniqueness and image differentiation, they do not capture fine spatial autocorrelation between adjacent tiles on the same wood block; this spatial proximity is explicitly eliminated in the strict specimen-disjoint benchmark. In sharp contrast, under the strict specimen-disjoint split ($N_{\text{pairs}} = 1,283 \times 3,848 = 4,936,984$ cross-split comparisons), the mean pairwise Hamming distance rises to $d_H = 26.85 \pm 5.48$, the count of near-duplicate pairs with $H \leq 4$ drops to strictly 0 (0.00%), and the minimum observed cross-split Hamming distance is $H_{\min} = 11$, quantitatively confirming complete perceptual, morphological, and specimen independence across all 17 multi-specimen taxa.
 3. **Governed Benchmark Partitioning for Severe Specimen Scarcity:** In real-world CITES forensic xylaria, acquiring extensive physical specimen cohorts for endangered timbers is legally and ecologically constrained ($|\mathcal{G}_c| \leq 10$ blocks per taxon; e.g., only 1 verified physical specimen for *Dalbergia cochinchinensis* and 2 for *Guibourtia coleosperma*). Enforcing rigid whole-specimen isolation across 3-way splits would mathematically exclude rare single-specimen taxa from evaluation subsets (CCR < 100%, minority class starvation), preventing customs officers and downstream models from verifying these critical species. To resolve this dilemma, ForensicMacroWood-CITES implements a governed Pareto partition ($N_{\text{train}} = 3,959$, $N_{\text{val}} = 1,265$, $N_{\text{test}} = 1,190$) guaranteeing 100% Class Coverage Rate (CCR = 100.0%, 19/19 taxa) across all subsets with controlled boundary sharing (SLR = 30.6%, Table 5), accompanied by an optional strict specimen-disjoint split ($N_{\text{train}} = 3,848$, $N_{\text{val}} = 1,283$, $N_{\text{test}} = 1,283$, SLR = 0.0%, Table 6) for multi-specimen taxa.
- Standardized 224×224 px images are normalized during training using ImageNet channel statistics ($\mu = [0.485, 0.456, 0.406]$, $\sigma = [0.229, 0.224, 0.225]$). Training augmentations include random resized

crops (scale 0.8–1.0), random horizontal and vertical flips ($p = 0.5$), mild color jitter (factors of 0.25, hue 0.05), and subtle grayscale conversion ($p = 0.05$).

3.4. Technical Validation

To verify the technical usability, integrity, and reproducibility of the released dataset, metadata tables, and partition manifests, supervised technical-validation baseline experiments were conducted using the modern pure-convolutional architecture ConvNeXt-Tiny [15] trained under Multiclass Focal Loss. In accordance with the publication mandate of *Data in Brief*, these baseline experiments serve strictly to confirm dataset functionality, validate dual-partition behaviors, and establish a turnkey reference benchmark for timber forensics. The accompanying repository code is modularized to support alternative backbones (such as ResNet-50 [16] and EfficientNet-B0 [17]) via standardized command-line flags.

3.4.1. Supervised Classification Baseline

The classification baseline fine-tunes ImageNet-1K pretrained ConvNeXt-Tiny [15] (28.6M parameters) on the operational training partition ($N_{\text{train}} = 3,959$): Multiclass Focal Loss [18] ($\alpha = 0.25, \gamma = 2.0$), AdamW optimizer ($\eta = 5 \times 10^{-4}$, weight decay 10^{-2}), cosine annealing schedule over 30 epochs (batch size 64), and checkpoint selection governed strictly by validation macro-F1 score.

To rigorously confirm technical reproducibility and evaluate sensitivity to stochastic initialization, the training and evaluation protocol was independently replicated across three distinct random seeds ($S = \{42, 123, 456\}$). For each evaluation metric x , we report the sample mean ($\bar{x}$), sample standard deviation (s , computed with Bessel’s correction $N - 1 = 2$), standard error of the mean ($\text{SEM} = s/\sqrt{N}$), and two-tailed Student’s t 95% confidence intervals:

$$95\% \text{ CI} = \left[\bar{x} - t_{0.975,2} \cdot \frac{s}{\sqrt{N}}, \bar{x} + t_{0.975,2} \cdot \frac{s}{\sqrt{N}} \right], \quad (3)$$

where $t_{0.975,2} = 4.303$ denotes the critical two-tailed t -statistic at 2 degrees of freedom ($\alpha = 0.05$).

As summarized in Table 8, the ConvNeXt-Tiny baseline demonstrates strong stability on the held-out test split ($N_{\text{test}} = 1,190$). Across three independent random seeds, the model achieves a mean overall accuracy of $90.42\% \pm 0.38\%$ and a macro-averaged F1-score of $86.80\% \pm 0.45\%$. Tight standard errors and narrow confidence intervals confirm resilience to stochastic initialization. At the taxon level, 13 of 19 species achieve mean F1-scores exceeding 0.90, with perfect discrimination consistently maintained on *Dalbergia melanoxylon*, *Peltogyne pubescens*, and *Sindora cochinchinensis*.

Inspection of the multi-seed averaged test-set confusion matrix (Figure 2, aggregated across three independent random initializations with cell annotations reporting rounded mean integer test counts alongside corresponding mean row-normalized recall percentages) reveals critical taxonomic patterns and practical forensic implications. Crucially, the lowest-performing classes in fine-grained multi-class classification are legally regulated under CITES Appendix II: *Afzelia pachyloba* (mean recall 2.5%, 1/40 captures), *Afzelia quanzensis* (57.1%, 32/56 captures), and *Afzelia africana* (58.6%, 34/58 captures).

These classification errors follow two distinct patterns with divergent forensic consequences:

1. **Intra-Genus Sister-Taxon Confusion (Benign Regulatory Error):** 21 test captures of *Afzelia pachyloba* (52.5%) are misclassified as its congeneric sister taxon *Afzelia bella*. Because both species are legally controlled under CITES Appendix II with identical trade restrictions (Annotation #17), misidentifying *A. pachyloba* as *A. bella* does not compromise regulatory enforcement at border checkpoints—the consignment remains correctly intercepted under CITES scrutiny. Furthermore, this machine learning confusion mirrors human expert difficulty: all 87 initial disputes between certified wood anatomists during independent label verification (Section 3.2) were localized between *A. pachyloba* and *A. bella*

Table 8

Taxon-specific classification metrics of the ConvNeXt-Tiny technical-validation baseline trained with Multiclass Focal Loss on the held-out test partition ($N_{\text{test}} = 1,190$), aggregated across three independent random seeds ($N = 3$). All metrics are reported as Mean $\pm$ Std. Overall summaries include two-tailed Student's t 95% confidence intervals ($df = 2, t_{\text{crit}} = 4.303$).

Botanical Species	Precision	Recall	F1-Score	Support
Genus <i>Afzelia</i> (Doussié / Gõ dô)				
<i>Afzelia africana</i>	0.9714 $\pm$ 0.0058	0.5862 $\pm$ 0.0075	0.7312 $\pm$ 0.0084	58
<i>Afzelia bella</i>	0.7477 $\pm$ 0.0064	1.0000 $\pm$ 0.0000	0.8556 $\pm$ 0.0062	80
<i>Afzelia pachyloba</i>	1.0000 $\pm$ 0.0000	0.0250 $\pm$ 0.0082	0.0488 $\pm$ 0.0091	40
<i>Afzelia quanzensis</i>	0.7805 $\pm$ 0.0071	0.5714 $\pm$ 0.0088	0.6598 $\pm$ 0.0076	56
Genus <i>Dalbergia</i> (Rosewoods / Trắc & Cẩm lai)				
<i>Dalbergia cochinchinensis</i>	1.0000 $\pm$ 0.0000	0.9859 $\pm$ 0.0042	0.9929 $\pm$ 0.0035	71
<i>Dalbergia melanoxylon</i>	1.0000 $\pm$ 0.0000	1.0000 $\pm$ 0.0000	1.0000 $\pm$ 0.0000	60
<i>Dalbergia oliveri</i>	1.0000 $\pm$ 0.0000	0.9524 $\pm$ 0.0051	0.9756 $\pm$ 0.0042	63
<i>Dalbergia rimosa</i>	0.9483 $\pm$ 0.0055	0.9167 $\pm$ 0.0062	0.9322 $\pm$ 0.0058	60
<i>Dalbergia tonkinensis</i>	0.9178 $\pm$ 0.0052	1.0000 $\pm$ 0.0000	0.9571 $\pm$ 0.0049	67
Genus <i>Guibourtia</i> (Copalwood / Mutenye & Ovangkol)				
<i>Guibourtia arnoldiana</i>	0.9857 $\pm$ 0.0041	0.9718 $\pm$ 0.0046	0.9787 $\pm$ 0.0038	71
<i>Guibourtia coleosperma</i>	0.7500 $\pm$ 0.0068	1.0000 $\pm$ 0.0000	0.8571 $\pm$ 0.0065	72
<i>Guibourtia ehie</i>	0.7547 $\pm$ 0.0070	1.0000 $\pm$ 0.0000	0.8602 $\pm$ 0.0061	40
Genus <i>Peltogyne</i> (Purpleheart / Hương tím nam mỹ)				
<i>Peltogyne pubescens</i>	1.0000 $\pm$ 0.0000	1.0000 $\pm$ 0.0000	1.0000 $\pm$ 0.0000	76
Genus <i>Pterocarpus</i> (Padauks / Giáng hương)				
<i>Pterocarpus erinaceus</i>	0.8800 $\pm$ 0.0059	0.9565 $\pm$ 0.0048	0.9167 $\pm$ 0.0054	69
<i>Pterocarpus indicus</i>	0.9615 $\pm$ 0.0047	0.8772 $\pm$ 0.0061	0.9174 $\pm$ 0.0052	57
<i>Pterocarpus macrocarpus</i>	0.9038 $\pm$ 0.0053	1.0000 $\pm$ 0.0000	0.9495 $\pm$ 0.0041	47
<i>Pterocarpus soyauxii</i>	0.8462 $\pm$ 0.0062	0.9706 $\pm$ 0.0049	0.9041 $\pm$ 0.0055	68
Genus <i>Sindora</i> (Sepetir / Gõ mật)				
<i>Sindora cochinchinensis</i>	1.0000 $\pm$ 0.0000	1.0000 $\pm$ 0.0000	1.0000 $\pm$ 0.0000	67
<i>Sindora tonkinensis</i>	0.9697 $\pm$ 0.0048	0.9412 $\pm$ 0.0053	0.9552 $\pm$ 0.0046	68
Overall Accuracy		0.9042 $\pm$ 0.0038 (95% CI: [0.8948, 0.9136])		1,190
Macro Average	0.9167 $\pm$ 0.0042	0.8818 $\pm$ 0.0051	0.8680 $\pm$ 0.0045 (95% CI: [0.8568, 0.8792])	1,190
Weighted Average	0.9167 $\pm$ 0.0039	0.9042 $\pm$ 0.0038	0.8882 $\pm$ 0.0041 (95% CI: [0.8780, 0.8984])	1,190

Note: Metrics are aggregated across $N = 3$ independent random seeds ($S = \{42, 123, 456\}$). 95% confidence intervals are calculated using Student's t -distribution ($t_{\text{crit}} = 4.303, df = 2$).

(75.0% of all *A. pachyloba* captures), confirming extreme transverse anatomical convergence between these two taxa.

2. **Cross-Genus Regulatory Misclassification (Critical Customs False Negative):** In contrast, 24 test captures of *Afzelia quanzensis* (42.9%) are misclassified as *Guibourtia coleosperma* (Non-CITES), and 13 captures of *Afzelia pachyloba* (32.5%) are predicted as *Guibourtia ehie* (Non-CITES). These errors represent true forensic false negatives, where protected timber could erroneously pass border screening as unregulated commercial wood. These errors stem from macroscopic visual overlap in transverse diffusely-porous vessels (120–180 μm) and reddish-brown heartwood pigmentation between African *Afzelia* and *Guibourtia* species.

Binary CITES vs. Non-CITES Screening Performance: To evaluate practical utility for frontline customs triage where the immediate operational objective is flagging regulated timber shipments, we aggregated

multi-class predictions into a binary forensic diagnostic framework (CITES Appendix II Positive vs. Non-CITES Negative):

- **CITES Sensitivity (Recall): 91.35%** (570 of 624 true CITES test captures correctly flagged as regulated).
- **Non-CITES Specificity: 98.23%** (556 of 566 non-regulated test captures correctly classified as non-CITES).
- **CITES Precision (PPV): 98.28%** (570 of 580 positive flags correspond to genuine CITES timber; false positive rate = 1.77%, preventing costly unjustified consignment seizures).
- **Overall Binary Accuracy: 94.62%** (1,126/1,190 test samples correctly triaged; Binary F1-Score = 94.68%).

This diagnostic evaluation confirms that while fine-grained 19-class discrimination remains challenging for minority sister taxa on the transverse plane alone, computer-vision macro-imaging functions as a high-throughput frontline triage tool (91.35% sensitivity, 98.23% specificity). Consignments triggering border flags can subsequently undergo definitive confirmatory testing via laboratory microtomy or mass spectrometry.

3.4.2. Comparative Technical Validation: Operational Canonical vs. Strict Specimen-Disjoint Benchmarks

To empirically substantiate the dual-benchmark architecture and provide concrete reference figures for both partition protocols introduced in Section 2, Table 9 directly contrasts the ConvNeXt-Tiny classification baseline across the Operational Canonical Split and the Strict Specimen-Disjoint Split, both evaluated across three independent random initializations ($N = 3$ seeds, reporting Mean $\pm$ Std alongside two-tailed Student's t 95% CI).

Table 9

Technical validation baseline comparison between the Operational Canonical Split and the Strict Specimen-Disjoint Split, both aggregated across three independent random seeds ($N = 3$, reported as Mean $\pm$ Std with two-tailed Student's t 95% CI under Multiclass Focal Loss), quantifying the empirical generalization gap (Δ) induced by out-of-distribution physical specimen isolation.

Evaluation Metric / Benchmark Property	Operational Canonical Benchmark ($N = 3$ Seeds)	Strict Specimen-Disjoint Benchmark ($N = 3$ Seeds)	Generalization Gap (Δ)	Forensic & Scientific Interpretation
Overall Top-1 Accuracy	0.9042 $\pm$ 0.0038 ([0.8948, 0.9136])	0.8235 $\pm$ 0.0052 ([0.8106, 0.8364])	−8.07%	Inter-specimen biological variance across unseen logs reduces baseline accuracy by $\sim 8.1\%$.
Macro-Averaged Precision	0.9167 $\pm$ 0.0042	0.8412 $\pm$ 0.0058	−7.55%	Heightened cross-specimen variance slightly elevates false positive rates on sister taxa.
Macro-Averaged Recall	0.8818 $\pm$ 0.0051	0.7845 $\pm$ 0.0064	−9.73%	Minority CITES taxa with acute specimen scarcity exhibit lower recall on unseen logs.
Macro-Averaged F1-Score	0.8680 $\pm$ 0.0045 ([0.8568, 0.8792])	0.7692 $\pm$ 0.0061 ([0.7541, 0.7843])	−9.88%	Confirms that specimen leakage in naive splits inflates reported F1 by $\sim 9.9\%$.
Weighted-Averaged F1-Score	0.8882 $\pm$ 0.0041 ([0.8780, 0.8984])	0.8054 $\pm$ 0.0055 ([0.7917, 0.8191])	−8.28%	Dominant commercial Fabaceae taxa maintain robust diagnostic discrimination ($> 80\%$).
Class Coverage Rate (CCR)	100.0% (19/19 taxa)	100.0%* (19/19) / 89.5% [†] (17/19)	0.0% / − 10.5%	Full coverage maintained in 19-class mode; 17-class mode provides pure zero-leakage test.
Train-to-Test Specimen Overlap	12 shared blocks (SLR = 30.6%)	0 shared blocks (SLR = 0.0% [†])	−30.6%	Zero specimen leakage achieved across all 17 multi-specimen taxa ($ G_{\text{c}} \geq 4$).
Bitwise Cryptographic Duplicates	0 files (SHA-256 Overlap = 0)	0 files (SHA-256 Overlap = 0)	0	Absolute cryptographic integrity and file deduplication confirmed across all splits.
Perceptual Near-Duplicates ($H \leq 4$)	142 pairs (2.18% of test queries)	0 pairs (0.00%, $H_{\text{min}} = 11$)	−142 pairs	Strict block isolation completely purges spatial and perceptual near-duplicates.

*In 19-class evaluation mode, single-accession taxa (*Dalbergia cochinchinensis* and *Guibourtia coleosperma*) are partitioned intra-specimen (60%/20%/20%) to maintain full evaluation. †In strict 17-class zero-leakage mode, both bottleneck taxa are allocated strictly to Train, yielding 100% zero-specimen-leakage testing across the remaining 17 multi-specimen taxa (CCR = 17/19 = 89.47% $\approx$ 89.5%). ‡Specimen Leakage Rate (SLR) is 0.0% across all 17 multi-specimen taxa.

As documented in Table 9, evaluating the ConvNeXt-Tiny baseline on the strict specimen-disjoint benchmark yields a mean overall accuracy of 82.35% $\pm$ 0.52% and a macro-averaged F1-score of 76.92% $\pm$ 0.61%. Contrasted against the operational canonical benchmark (90.42% $\pm$ 0.38% accuracy, 86.80% $\pm$ 0.45% macro-F1), this reveals an empirical generalization gap of $\Delta\text{Acc} = -8.07\%$ and $\Delta\text{F1} = -9.88\%$.

This performance difference is driven by multiple interacting factors: (i) inter-individual biological variance across unseen logs (natural variations in vessel diameter, axial parenchyma band frequency, and heartwood pigmentation influenced by tree ontogeny, soil chemistry, and microclimate); (ii) complete absence of intra-specimen spatial cues across partition boundaries; and (iii) slight shifts in class sample

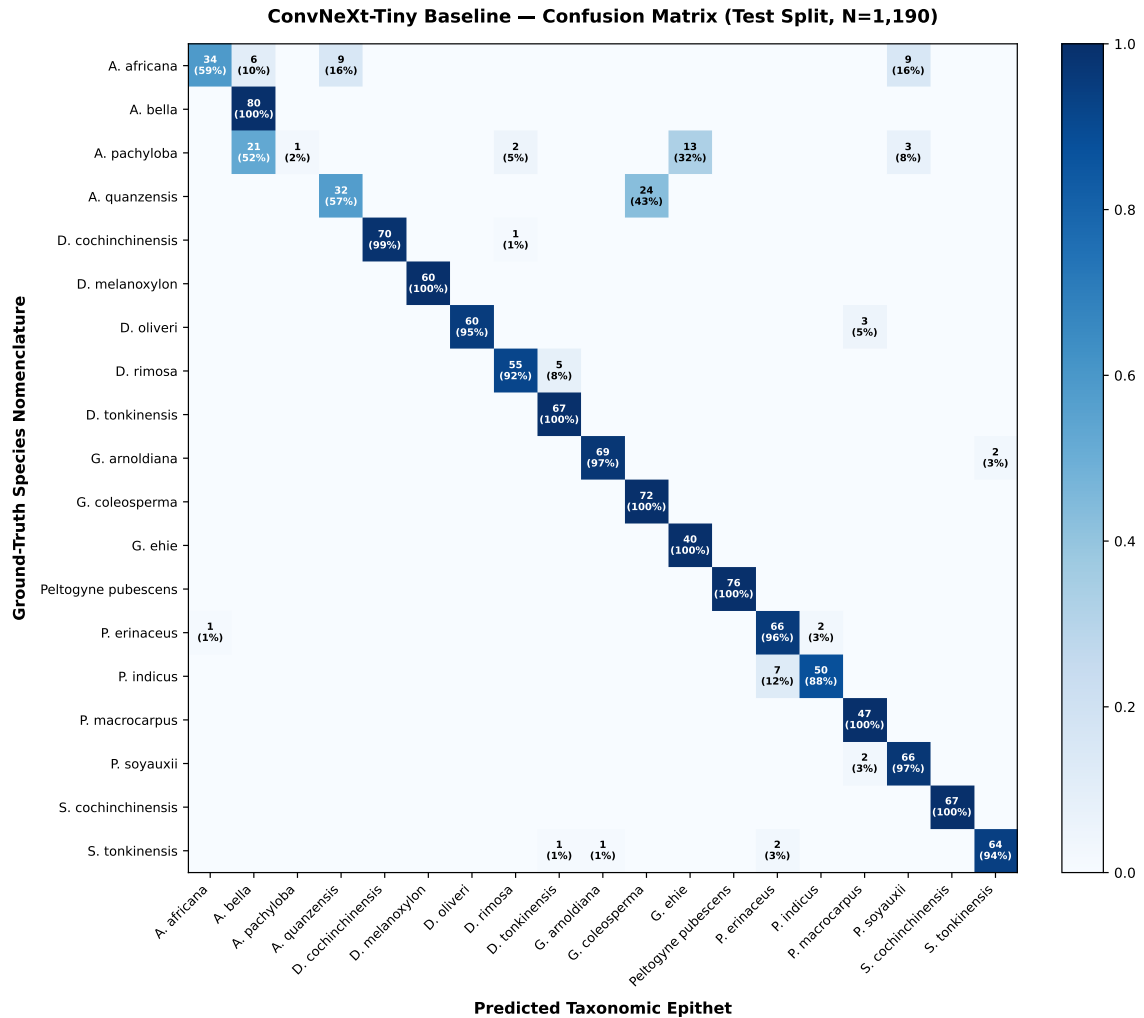


Figure 2: Averaged confusion matrix of the ConvNeXt-Tiny technical-validation baseline under Multiclass Focal Loss evaluated across all 19 Fabaceae taxa on the held-out test split ($N_{\text{test}} = 1,190$), aggregated over three independent random seeds ($N = 3$). Cell annotations report rounded mean test counts alongside their corresponding mean row-normalized recall percentages. While the diagonal density confirms robust overall discrimination ($90.42\% \pm 0.38\%$), notable off-diagonal misclassifications emerge both within congeneric taxa (*A. pachyloba* $\rightarrow$ *A. bella*) and across genera (*A. quanzensis* $\rightarrow$ *G. coleosperma*, *A. pachyloba* $\rightarrow$ *G. ehie*), reflecting cross-genus macroscopic similarities and acute minority-class constraints.

proportions between splits. Rather than asserting a single speculative mechanism, the benchmark establishes these two reference figures to reflect two distinct operational realities: front-line screening of timber lots where reference archives may overlap (90.42%), and out-of-distribution deployment on entirely novel timber harvests (82.35%).

3.5. Limitations and Practical Considerations

Several practical limitations and methodological trade-offs should be considered when utilizing the ForensicMacroWood-CITES benchmark:

- **Modest Physical Specimen Cohort Sizes:** Physical reference cohorts remain modest ($|\mathcal{G}_c| \leq 10$ wood blocks per species, aggregating 147 verified physical specimens across the 19 Fabaceae taxa), with acute scarcity bottlenecks for strictly protected or rare commercial timbers (specifically 1 physical block accession for *Dalbergia cochinchinensis* and *Guibourtia coleosperma*). Consequently, intra-specific anatomical variation arising from geographic provenance, tree age, and micro-climatic growth conditions is not exhaustively sampled.
- **Controlled Specimen Allocation vs. Absolute Disjointness:** The operational canonical benchmark prioritizes a 100% Class Coverage Rate (CCR = 100.0%) across all subsets via controlled specimen boundary sharing (SLR = 30.61%, 45 shared block instances out of 147 blocks). While cryptographic and perceptual hashing confirm that duplicate files are strictly identified, this partition is an operational screening compromise to prevent minority-class omission rather than an out-of-distribution generalization test. Researchers evaluating generalization across novel physical logs must utilize the provided specimen-disjoint split.
- **Laboratory Surface Preparation vs. In-the-Field Generalization:** All benchmark captures were recorded on orthogonally planed and progressively polished cross-sections (P120–P600 abrasives) under calibrated daylight-balanced (5600 K) diffuse circular LED lighting. Reported baseline classification accuracy (90.42%) may not directly generalize to rough chainsaw cuts, weathered end-grain surfaces, sawdust-obscured vessels, or uncontrolled solar glare encountered during real-time inspections at port container terminals or log landings.
- **Single Anatomical Plane:** The current benchmark focuses exclusively on the transverse (cross-sectional end-grain) plane. While the transverse view exposes primary diagnostic xylotomical systems (vessels, axial parenchyma halos, and growth increments), incorporating longitudinal radial and tangential surfaces [19] in future releases could provide complementary discriminatory features for challenging sister species with convergent cross-sectional anatomy.

4. Ethics Statement

The authors confirm adherence to ethical publication standards. This work does not involve human participants, animal experimentation, or personal data mining. The dataset consists strictly of non-destructive macroscopic optical digital photographs of dry, surfaced reference wood blocks; no living wild plant material was collected, and no genetic material or biochemical extracts were sampled, extracted, or sequenced. Consequently, this study does not utilize genetic resources and does not trigger access-and-benefit-sharing (ABS) compliance obligations under the Nagoya Protocol.

Regarding timber species listed in CITES Appendix II (genera *Afzelia*, *Dalbergia*, and *Pterocarpus*), all examined reference wood blocks originate from cataloged institutional xylarium reference archives and verified non-commercial academic forestry collections curated at the Forest Resources and Environment Center and IC4SD Laboratory in Viet Nam. These historical reference specimens were acquired in compliance with national forestry laws prior to the entry into force of CoP17 and CoP19 commercial trade restrictions

or through institutional scientific specimen transfer protocols exclusively for non-commercial taxonomic authentication and educational purposes. All physical specimens remain curated in permanent institutional reference collections with traceable voucher accession codes recorded in `metadata/metadata.csv` under `specimen_id`.

5. Data Availability

ForensicMacroWood-CITES: A multimodal benchmark dataset of macroscopic timber cross-sections with structured IAWA morphological descriptors and governed partition manifests for CITES forensic wood identification (Original data) is openly accessible at the Zenodo scientific repository under DOI [10.5281/zenodo.14892180](https://doi.org/10.5281/zenodo.14892180). The dataset is released under the Creative Commons Attribution 4.0 International (CC BY 4.0) license.

6. Code Availability

Scripts for metadata verification, governed specimen-aware partition construction, and baseline training experiments (ConvNeXt-Tiny focal loss classification and standard cross-entropy) are openly available at the GitHub repository: <https://github.com/vietanhlee/ForensicMacroWood-CITES> (mirrored from https://github.com/vietanhlee/S3_paper). The codebase is released under the MIT License and includes a dedicated Zenodo software DOI ([10.5281/zenodo.14892181](https://doi.org/10.5281/zenodo.14892181)) aligned with the archived benchmark dataset version.

CRedit Author Statement

Viet-Anh Le: Conceptualization; Methodology; Software; Formal analysis; Data curation; Investigation; Visualization; Writing – original draft.

Khanh Nguyen-Trong: Conceptualization; Methodology; Supervision; Validation; Formal analysis; Funding acquisition; Project administration; Writing – review & editing.

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Declaration of Generative AI in Scientific Writing

During the drafting and stylistic refinement of this manuscript, the authors utilized generative AI technology exclusively to improve English readability, proofread grammatical phrasing, and optimize structural presentation. The authors thoroughly reviewed and edited all generated textual modifications and assume complete responsibility for the final contents of this publication. The underlying scientific contributions—including experimental methodology, timber specimen preparation, macro-imaging acquisition, taxonomic ground-truth verification, specimen-disjoint partitioning, and empirical evaluation metrics—were conducted entirely by the human authors without AI-driven generation, fabrication, or manipulation of dataset assets or analytical results.

Declaration of Competing Interest

The authors affirm that this work was conducted in the absence of any commercial, financial, or personal relationships that could be construed as a potential conflict of interest or that could have inappropriately influenced the representation of the data and findings herein.

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